UNIFIED MECHANICS

A WORLD OF DISTINCTIONS

The Complete Physical and Mathematical Reconstruction

Observation first. Mechanism made explicit. Mathematics as formal compression.

FINAL TWO-PAPER EDITION | MAIN MONOGRAPH

This edition consolidates the physical framework, the mathematical reconstruction, the observational language, the multiverse constraint algebra, and the empirical programme into one dependency-ordered account. The physical framework is the source semantics. The mathematics formalises, compresses, and tests it.

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THE GOVERNING SENTENCE

A realised system retains identity through distinctions, responds to local conditions through admissible processes, and describes those processes by the records they leave. Unified Mechanics is the attempt to write that same operation once in physical language and once in mathematics, without mistaking either language for the system itself.

This edition is deliberately written in two orders at once. The discovery order begins with the physical picture: systems propagate, accumulate, meet at boundaries, stabilise around anchors, fail into denser anchor states, and leave records. The formal order asks what minimal mathematical structure reproduces those same relations. The two orders meet at an explicit realisation map. Neither is permitted to silently replace the other.

> *The framework did not begin by choosing the golden ratio and inventing a universe around it. It began with a physical account of how a universe maintains itself. The mathematics was built to compress that account.*
>
> **Editorial rule of the final edition**

Abstract

Unified Mechanics is presented here as one systems framework with three inseparable layers. The first is a physical ontology of realised systems: distinction, retained identity, relation, boundary, local condition, transformation, record, and observation. The second is an internal mathematical reconstruction: normal forms, semantic quotient, whole-part-remainder closure, a two-rate partition, the golden fixed point, and Primordial Loop Algebra. The third is the physical realisation: light as propagation, matter as accumulation, boundary as the interface and constraint channel, gravity as boundary response, black holes as saturated anchor states, and adjacent recursion levels as persistent boundary conditions. The document derives the metallic-ratio constraint ladder from two primitive projective operations, distinguishes the full loop algebra from its primitive one-inversion spine, and corrects the ontology of the accumulator, the n=0 mirror state, the first attracting golden universe, and the silver child. It then constructs a zero-fit gold-silver interface acceleration scale and a minimal bilinear galaxy closure that produces flat rotation, the baryonic fourth-power relation, an apparent linearly growing halo mass, and a $1/R^2$ effective density. Cosmological, quantum, particle, thermodynamic, semantic, solver, and empirical claims are gathered into one graded ledger. Every physical claim is paired with a mechanism, a record channel, and a falsification surface.

Reading protocol and claim grades

DO NOT COLLAPSE THE GRADES

A mathematical theorem, a physical realisation, a proposed constitutive law, and an empirical result are different kinds of statement. This monograph keeps them in one chain but never treats them as interchangeable.

Grade	Meaning
PROVED	Follows inside the stated mathematical definitions and assumptions.
CONDITIONAL	Follows once an explicitly named representation or physical realisation condition is accepted.
CONJECTURED	A precise extension suggested by the framework but not yet closed by proof or decisive test.
PROPOSED	A concrete physical model or constitutive law offered for direct calculation and falsification.
POSTULATE	A primitive physical or semantic commitment used to connect the internal mathematics to realised systems.

Observed data and executed computations are reported as records rather than elevated to a sixth logical grade. They can support or falsify a proposed realisation; they do not change the truth of an internal theorem.

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Part I - The physical source model

DISCOVERY ORDER

The physics in this Part is not presented as a metaphor added after the equations. It is the source model that the later mathematics is required to encode. The historical simulations and early physical papers belong here, before the formal closure.

1. A world of distinctions

A system is not first recognised by a number. It is recognised because something remains distinguishable while something changes. A cup remains the same cup while it is moved. A water molecule remains H_2O while it rotates, vibrates, collides, and changes local environment. A galaxy remains one organised system while stars orbit, gas cools, matter accretes, and records arrive at an observer. Retained identity and admissible change are therefore prior to any particular coordinate description.

POSTULATE - REALISATION

A physical description requires a nonempty realised carrier. Mathematics may describe an empty case, but it does not create a physical carrier merely by describing one.

The null system is useful precisely because it marks the boundary. Let N denote the absence of a realised carrier. N has no retained whole, no parts, no local conditions, no process, and no record. Its complete description is therefore not a hidden physical vacuum; it is the instruction that the physical clauses of the ontology do not execute.

$$N = (\varnothing \text{ whole}, \varnothing \text{ parts}, \varnothing \text{ boundary}, \varnothing \text{ process}, \varnothing \text{ record}) \quad (1.1)$$

A realised system S begins when a carrier supports at least one distinction that persists long enough to constrain a later state. The minimum ontology is not a bag of nouns. It is an ordered relation among a retained whole, its parts, its boundary, its local conditions, its admissible transformations, and the records by which those transformations become available to another system.

Component	Operational meaning
Whole	The state or organisation treated as one retained system.
Parts	Nested components or roles whose relations realise the whole.
Boundary	The interface at which inside/outside, prior/later, or one mode/another is resolved.
Local conditions	The actual environment and state variables that constrain what the system can do here and now.
Admissible process	A transformation that preserves or deliberately changes the relevant identity conditions.
Record	A persistent physical difference from which a process can be reconstructed or inferred.

THE DECISIVE DISTINCTION

Description is not creation. A description is physically meaningful when a realised system changes another realised system in a way that preserves a record. The same sentence can be true as mathematics and unrealised as physics.

1.1 Distinction, identity, and relation

Distinction is the operation that prevents all content from collapsing into one undifferentiated statement. Identity is the retention of a distinction across an admissible change. Relation is the constraint that makes one distinction relevant to another. These are not three independent substances. They are three roles in one act of self-maintenance.

$$\text{realised system} = \text{retained distinction} + \text{admissible relation} + \text{recorded change} \quad (1.2)$$

The phrase "a world of distinctions" therefore does not claim that reality is made of labels. It says that every physically available fact already contains a separation, a persistence condition, and a relation through which that separation matters.

2. Realisation, local condition, and record

LOCAL-CONDITION REQUIREMENT

Identity never determines behaviour by itself. A complete physical description includes the local conditions that select which of the system’s admissible processes is realised.

H_2O is the clean example. Molecular identity fixes two hydrogen nuclei, one oxygen nucleus, the permitted electronic structure, and a characteristic relation among them. It does not by itself determine whether the realised system is ice, liquid, vapour, supercooled water, an isolated molecule, or a participant in a reaction. Temperature, pressure, neighbouring fields, boundary conditions, phase history, and measurement context are not optional decorations. They select the physical trajectory.

Layer	H_2O example
Retained identity	Two H nuclei and one O nucleus in the water molecular relation.
Local condition	Temperature, pressure, field environment, neighbours, phase boundary, and history.
Admissible process	Rotation, vibration, translation, dissociation, reaction, condensation, freezing.
Record	Spectrum, momentum transfer, phase boundary, chemical product, detector trace.

This is why a formula does not become an observation merely by producing the right number. The complete measurement chain is:

$$\text{mechanism} \rightarrow \text{coupling} \rightarrow \text{physical change} \rightarrow \text{record} \rightarrow \text{inference} \rightarrow \text{reported observable} \quad (2.1)$$

NO NUMBER WITHOUT A RECORD

Every numerical claim in Unified Mechanics must answer six questions: What system changes? Through what coupling? Under what local conditions? What record is produced? How is that record reconstructed? Which observable is finally reported?

2.1 The local self-description principle

A realised system describes itself operationally when its retained identity and relevant local conditions alter its boundary, interactions, or records so that a compatible receiver can reconstruct a state difference. Self-description is therefore not a system speaking English about itself. It is the unavoidable fact that what a system is constrains what it can do, and what it does leaves a physically structured trace.

$$D\_S : (\text{identity, local condition, process}) \mapsto \text{boundary effect} \mapsto \text{record} \quad (2.2)$$

A description may be lossy, delayed, frame-dependent, or inaccessible to a particular receiver. It remains a description if the record is generated by the system's own retained conditions rather than arbitrarily attached from outside.

3. Universopedia and the Familiarity Kernel

Universopedia is the observational-language layer of the framework. It begins with familiar realised systems rather than with symbols. For each system it separates direct observation, inference, relevant local condition, operative boundary, transformation, record, and uncertainty. It then compresses the system into the shared grammar without erasing the physical details that made the system recognisable.

FAMILIARITY KERNEL

A record enters the Familiarity Kernel when it is recognisable as structured but has not yet been assigned to a validated system class. The kernel preserves the record, its context, and competing reconstructions until a stable classification survives transfer and reconstruction.

Universopedia field	Question
Observed carrier	What physically exists or occurred?
Retained identity	What remained the same across the observation?
Parts and roles	Which components and relations realise that identity?
Local conditions	Which conditions selected this behaviour?
Boundary	Where was a distinction or transfer resolved?
Transformation	What changed?
Record	What persistent difference was measured?
Inference	What was reconstructed from the record?
Competing description	What else could generate the same record?

Universopedia field	Question
Status	Observed, inferred, conditional, proposed, or unresolved?

The compiler and the observational language therefore perform opposite but complementary tasks. The compiler removes redundant representation while preserving operational structure. Universopedia restores the physical carrier, local condition, and record needed to prevent an abstract normal form from being mistaken for a measurement.

4. The observed three-mode system

Across the early physical framework, three behaviours recur before the final algebra is introduced. Some structure propagates: it carries state outward and remains available across distance. Some structure accumulates: it retains, binds, clusters, and builds persistent local organisation. Whenever propagating and accumulating structures meet, an interface appears. The interface is not a third object seeded independently. It is the cross-relation forced by the coexistence of the other two behaviours.

Observed mode	Physical character	Later formal name
Propagation	Open transmission, exploration, signal carriage, geometric reporting.	Light channel
Interface	Acknowledgement, conversion, constraint, tension, inside/outside resolution.	Boundary channel
Accumulation	Retention, binding, clustering, familiarity, persistent structure.	Matter channel

SOURCE SEMANTICS

The names Light, Boundary, and Matter were not selected because three algebraic terms happened to be available. They name the physical operations that the later three terms are required to reproduce.

The structure simulations explore this realisation on deliberately unrelated substrates: reaction-diffusion fields, strange attractors, hypergraphs, fractal growth, causal rewriting, sky maps, and large-scale-structure catalogues. They are not proofs that nature must instantiate the model. Their role is narrower and valuable: they show that the same three-role projection can be implemented consistently across distinct dynamical substrates, and that the boundary appears where propagation and accumulation mix.

5. Anchors, Strategy A, Strategy B, and black holes

The physical framework interprets gravity as an anchor operation. An anchor does not need to add a new outward push. It removes freedom by coupling a local content distribution to a larger constraint structure. The boundary records both sides of the relation: matter changes geometry, and the changed geometry constrains matter. This bidirectional constraint is the physical meaning later encoded by the boundary cross-term and the metric field equation.

STRATEGY A

A realised region maintains a stable identity around a dominant anchor. Perturbations inside its recovery basin are redistributed through propagation, boundary response, and accumulation until the region returns to a coherent organisation.

STRATEGY B

When the local configuration can no longer retain the Strategy A identity, the region resolves by becoming a denser anchor state. The failed organisation is not left undefined; it becomes the constraint structure around which a larger-scale Strategy A system can form.

Black holes are the saturated local Strategy B case. A galactic halo is the distributed boundary-anchor case. A supermassive black hole and its galaxy exhibit both strategies simultaneously: a compact saturated anchor nested inside an extended stable organisation. The multiverse picture later applies the same physical grammar one scale above the universe rather than inventing a different mechanism.

$$\text{Strategy A around Strategy B} = \text{stable content around retained boundary infrastructure} \quad (5.1)$$

This repeated organisation is the reason the old corpus speaks of black holes, dark halos, galaxy centres, stellar systems, and atomic centres in one structural vocabulary. The claim is not that their standard equations are identical. The claim is that each realises the same system role: retained content organised by an anchor and mediated through a boundary.

Part II - The internal mathematical reconstruction

FORMAL ORDER

From this point the document asks what minimal mathematics preserves the source model. Every result before the physical realisation chapter is internal mathematics. Physical claims return only when the representation map is stated.

6. Self-description and semantic normal form

A language capable of describing systems must preserve the differences that affect execution while removing differences that arise only from notation. Unified Mechanics therefore separates syntax from operational meaning. Rewrite rules reduce expressions; termination prevents endless descriptive drift; local confluence prevents different valid reduction orders from producing incompatible final forms; and the semantic quotient identifies expressions that execute the same system relation.

$$\text{expression} \rightarrow \text{terminating rewrites} \rightarrow \text{unique normal form} \rightarrow \text{semantic class} \quad (6.1)$$

The normal form is not the physical system. It is the smallest representation that preserves the system's operational distinctions. A physical realisation must later supply carrier, local condition, coupling, and record.

Formal object	Role
Distinction	Separates states or roles that cannot be interchanged without changing execution.
Identity	Defines what must be retained across an admissible rewrite.
Relation	Specifies how distinctions constrain one another.
Normal form	Canonical representative after all admissible reductions.
Semantic quotient	Class of representations with the same operational action.
Realisation map	Assignment from an internal class to a physical carrier and observables.

ANTI-CIRCULARITY RULE

A physical observation may motivate the formal vocabulary, and the formal vocabulary may compress the observation. Circularity begins only if empirical agreement is used as the proof of an internal theorem, or if an internal theorem is treated as a measurement without a realisation map.

7. Whole, part, remainder, and the golden closure

The physical source model treats a system as a retained whole realised through related parts. The minimal nontrivial self-description asks for a ratio that remains the same when the viewpoint moves one level: whole relative to retained part equals retained part relative to remainder. Let the remainder be one unit and the retained part be x units. The whole is x+1.

$$(x + 1)/x = x/1 \quad (7.1)$$

$$x^2 = x + 1 \quad (7.2)$$

$$\varphi = (1 + \sqrt{5})/2 \quad (7.3)$$

The negative root is algebraically real but cannot represent a positive whole-to-part magnitude in this realisation. The golden ratio is therefore not inserted because it is aesthetically familiar. It is the positive fixed point of the minimal one-part self-similar closure.

PROVED - POSITIVE CLOSURE UNIQUENESS

Among positive real ratios satisfying the minimal whole/part/remainder self-similarity condition with one indivisible remainder, φ is unique.

This derivation does not yet say that the universe must use this closure. It proves the internal ratio. The physical source model motivates why a universe that retains itself through nested whole-part relations is represented by it.

7.1 The contraction coordinate

The smaller-to-larger ratio in the golden closure is 1/φ. A realised boundary resolves two opposed orientations: inward/outward, retained/transmitted, or source/receiver. The directed contraction assigned to either orientation is therefore half of the reciprocal ratio:

$$2r = 1/\varphi \quad \Rightarrow \quad r = 1/(2\varphi) = (\sqrt{5} - 1)/4 \quad (7.4)$$

CONDITIONAL - BIDIRECTIONAL NORMALISATION

The factor 2 is not an arbitrary numerical repair. It is the normalisation required when one reciprocal whole-part relation is resolved into two directed boundary roles. Accepting that physical two-sided realisation fixes r.

8. The two-rate partition and the three channels

Once a system has a retained rate r and its complement 1-r, a complete two-role interaction is the square of their sum. The expansion contains exactly the two pure paths and the mixed path:

$$1 = [r + (1 - r)]^2 = r^2 + 2r(1 - r) + (1 - r)^2 \quad (8.1)$$

Term	Exact value	Operational reading
$W_M = r^2$	0.0954915028	Two retained steps: accumulation, binding, persistent local structure.
$W_B = 2r(1-r)$	0.4270509831	Mixed step in either order: interface, exchange, constraint, acknowledgement.

Term	Exact value	Operational reading
$W\_L = (1-r)^2$	0.4774575141	Two transmitted steps: propagation, exploration, geometric reporting.

PROVED - PARTITION

The three weights sum to unity identically. The physical names are a realisation of the algebraic path types, not an additional numerical fit.

$$W\_L / W\_M = 5 \quad (8.2)$$

The propagation-to-accumulation throughput ratio is exactly five.

The boundary term cannot be generated by a pure retained path or a pure transmitted path. It is the cross-term and therefore exists only where the two modes are jointly realised. This is the formal compression of the physical observation that the interface is emergent rather than seeded.

9. Dynamics, stability, time, and the traversal floor

The golden fixed point can be written in the contraction coordinate as a recurrence. Substituting $\varphi=1/(2r)$ into $\varphi=1+1/\varphi$ gives:

$$r_{k+1} = 1/(4r_k + 2) \quad (9.1)$$

Its physical fixed point is r. Linearisation gives the multiplier:

$$\mu = -4r^2 = -1/\varphi^2 \quad (9.2)$$

The negative sign gives alternating correction across the boundary; the magnitude $1/\varphi^2<1$ gives attraction. The forward recurrence is stable while the inverse recurrence is unstable. Temporal direction is therefore represented by the direction in which retained distinctions converge rather than amplify.

CONDITIONAL - ARROW OF TIME

When physical time is realised as ordered execution of the recurrence, the stable direction is future-directed and the inverse unstable direction is past-directed. The algebra supplies temporal orientation; a dimensional clock still requires a physical process.

9.1 The traversal floor

A complete recorded process must traverse the three operational roles: a retained difference must propagate, be resolved at a boundary, and be retained again as a record. If each unresolved transfer carries the contraction factor r, the residual of one complete three-role cycle is:

$$\varepsilon = r^3 = 0.0295084972\ldots \quad (9.3)$$

CONDITIONAL - FLOOR THEOREM

The cubic floor follows from multiplicative contraction across one Light-Boundary-Matter traversal. It is a physical process floor only under the three-role traversal realisation; it is not a universal excuse for arbitrary numerical residuals.

This corrected statement replaces the looser practice of treating every few-percent discrepancy as automatically acceptable. A claimed floor must identify the traversal count, the coupling path, and the record in which the residual appears.

Part III - Primordial Loop Algebra

PURPOSE OF PLA

The golden closure constrains our universe. Primordial Loop Algebra asks what constrains the family of possible closures. It is the rule layer above the single golden fixed point.

10. Projective system states and primitive edits

Represent a scale-free whole-part state by a primitive positive pair (u,v), modulo common scale. The physical magnitude is the projective ratio x=u/v. Two edits are inherited directly from the source semantics. A system can extend the retained whole by one indivisible part, or it can reverse the roles from which whole and part are read.

$$A : (u,v) \mapsto (u+v,v) \quad \Rightarrow \quad x \mapsto x+1 \qquad (10.1)$$

$$J : (u,v) \mapsto (v,u) \quad \Rightarrow \quad x \mapsto 1/x \qquad (10.2)$$

Primitive	Physical reading	Matrix
A - accumulate	Extend content without changing the unit part.	[[1,1],[0,1]]
J - invert	Exchange whole and part; reverse the descriptive viewpoint.	[[0,1],[1,0]]

A alone never closes at a finite positive fixed point. J alone fixes x=1 but supplies no growth. A realised nontrivial loop therefore requires both extension and reversal.

11. Completeness of accumulation and inversion

DEFINITION - ADMISSIBLE LATTICE EDIT

An admissible primitive edit acts on positive primitive integer pairs, preserves the integer lattice and scale-free projective meaning, and is unimodular so that no distinguishable lattice information is silently created or destroyed.

Every nonnegative unimodular 2×2 integer matrix can be reduced to the identity by the Euclidean algorithm: subtract the smaller column from the larger and, when required, exchange the columns. Reversing the reduction expresses the original map as a word in A and J. Thus A and J are complete for the positive scale-free lattice transformations admitted by the definition.

PROVED - POSITIVE COMPLETENESS

The monoid generated by A and J contains every nonnegative unimodular projective whole-part transformation. Allowing A^{-1} extends the construction to the integer projective group generated by translation and inversion.

The precise symmetry language is important. With positive words, PLA is a continued-fraction semigroup. With inverse translations admitted, its natural projective setting is PGL(2,Z), the extended modular lattice.

Calling the two positive generators the whole of PSL(2,R) obscures both the integer constraint and the orientation reversal carried by J.

12. The full algebra and the primitive metallic spine

A general loop word may contain several inversions:

$$W = A^{a1}\ J\ A^{a2}\ J \cdots A^{ak}\ J \quad (12.1)$$

Its fixed point is a periodic continued fraction and, for an admissible hyperbolic word, a quadratic irrational. The full PLA therefore contains far more than the metallic ratios. The old statement that accumulation and inversion directly produce only one one-dimensional metallic ladder was incomplete.

DEFINITION - PRIMITIVE UNIVERSE

A primitive universe is a closed loop with no proper inversion-delimited internal loop. Multiple inversion-delimited segments represent a composite braid whose subloops can already be described separately.

Every finite positive closed word must contain at least one inversion. If it contains more than one, the inversion-delimited segments define proper internal cycles and the word is composite under the definition. The primitive word therefore contains exactly one inversion:

$$M_n = A^nJ = [[n,1],[1,0]] \quad (12.2)$$

$$x = n + 1/x \quad \Rightarrow \quad x^2 - nx - 1 = 0 \quad (12.3)$$

$$\varphi_n = (n + \sqrt{(n^2+4)})/2 \quad (12.4)$$

PROVED - PRIMITIVE CLOSURE THEOREM

Under the stated minimality definition, the primitive positive closed loops are exactly the one-inversion words A^nJ. Their positive fixed points form the metallic-ratio spine. Composite periodic words remain valid PLA structures but are internal braids rather than new primitive rungs.

n	Name	φ_n	Status
0	mirror / unity	1	Algebraic fixed point; neutral under direct iteration.
1	golden	1.6180339887	First attracting primitive universe.
2	silver	2.4142135624	Next attracting rung; child and adjacent upper neighbour.
3	bronze	3.3027756377	Higher light-dominant attracting rung.

13. Stability and the corrected genesis sequence

For the fixed-point iteration $f_n(x)=n+1/x$, the derivative at φ_n is:

$$f_n'(\varphi_n) = -1/\varphi_n^2 \quad (13.1)$$

The distinction between a fixed point and an attractor resolves several old corpus collisions. At n=0, $|f'|=1$: the unity mirror is neutral, not an autonomous attracting universe. For n≥1, $|f'|<1$: the metallic rungs are attracting. The formal n=−1 solution requires $A^{-1}J$ and has $|f'|>1$ at its positive root; it is a repelling threshold or separatrix unless a different dynamics is explicitly supplied.

CORRECTED GENESIS ONTOLOGY

Π is the unclosed accumulating substrate. $A^{-1}J$ marks the repelling self-contact threshold. n=0 is the first closed neutral mirror. n=1 is the first attracting primitive universe. n=2 is its silver child and permanent adjacent upper boundary. The n→∞ closed-rung limit is light-dominant and must not be identified with the bare unclosed accumulator.

Stage	Algebraic object	Physical role
Unclosed substrate Π	A without a finite fixed point	Pre-closure accumulation; no channel partition assigned.
Self-contact threshold	$A^{-1}J$	Repelling separatrix; onset of role reversal.
Mirror closure	$J = A^0J$	Neutral boundary state with x=1.
Golden universe	AJ	First attracting self-consistent primitive closure.
Silver child	A^2J	Next attracting closure generated across the golden upper boundary.

This correction preserves the physical genesis story while preventing four distinct objects - substrate, threshold, mirror, and universe - from being called the same primordial state.

14. The multiverse constraint lattice

Each primitive rung is an integer-trace, determinant-minus-one matrix:

$$\det(M_n)=-1, \quad \mathrm{tr}(M_n)=n \quad (14.1)$$

The positive eigenvalue is φ_n and the second eigenvalue is $-1/\varphi_n$. Each rung therefore carries an expanding direction and a reversing contracting direction. The family is a discrete projective constraint lattice, not a set of freely chosen constants.

$$r_n = 1/(2\varphi_n) \quad (14.2)$$

$$W\_L^{(n)}=(1-r_n)^2, \;\; W\_B^{(n)}=2r_n(1-r_n), \;\; W\_M^{(n)}=r_n^2, \;\; \varepsilon_n=r_n^3 \quad (14.3)$$

n	Rung	φ_n	W_L	W_B	W_M	ε_n
0	mirror	1.000000	0.250000	0.500000	0.250000	12.500%
1	gold	1.618034	0.477458	0.427051	0.095492	2.951%
2	silver	2.414214	0.628680	0.328427	0.042893	0.888%
3	bronze	3.302776	0.720143	0.256939	0.022918	0.347%

EXACT CROSSING IDENTITY

The floor drop from the n=0 mirror to the n=1 golden universe equals the golden matter weight exactly: $1/8 - r^3 = r^2$. The matter channel is therefore the exact algebraic remainder of the first attracting closure from the neutral mirror.

$$\varepsilon_0 - \varepsilon_1 = 1/8 - r^3 = r^2 = W\_M \quad (14.4)$$

Part IV - Physical realisation

THE BRIDGE

The mathematics now has a complete internal chain. Physics returns by assigning its roles to realised carriers, couplings, local conditions, and records. From this point every result is conditional on the stated realisation unless separately observed.

15. The representation bridge

The representation principle maps the three formal paths to the physical source model already observed: transmitted-transmitted paths realise propagation; mixed paths realise the boundary interface; retained-retained paths realise accumulation. The map is justified by operational action, not by numerical resemblance.

$$R : (W\_L, W\_B, W\_M) \mapsto (\text{propagation, interface, accumulation}) \quad (15.1)$$

Formal structure	Physical realisation	Primary record
W_L pure complement path	Light/information propagation and geometric reporting	Photon, wave, phase, null trajectory, transmitted signal
W_B mixed path	Boundary exchange, gravity, constraint, conversion, holographic record	Curvature, redshift, lensing, interface stress, transition
W_M pure retained path	Matter, binding, memory, persistent local structure	Mass distribution, bound state, chemical or material record

The map is not one-to-one at the level of substances. A photon is a light-channel excitation, but a measured photon event also requires a boundary and a retained detector record. A galaxy is matter-rich locally, boundary-governed dynamically, and light-reported observationally. Channel dominance and complete process are different questions.

16. Boundary gravity and the field equation

The minimal metric realisation introduces a light-coupling or boundary-response field λ(x). The curvature term is weighted by the local ability of the realised system to transmit geometric information:

$$S = \int d^4x \, \sqrt{-g} \, [\, \lambda R/(16\pi G) - Z(\lambda)(\partial\lambda)^2/2 - U(\lambda) + L\_m \,] \quad (16.1)$$

Variation with respect to the metric gives:

$$\lambda G\_\{\mu\nu\} + (g\_\{\mu\nu\}\Box - \nabla\_\mu\nabla\_\nu)\lambda = 8\pi G(T\_\{\mu\nu\}+T\textasciicircum\{(\lambda)\}\_\{\mu\nu\}) \quad (16.2)$$

MECHANISM, NOT ANALOGY

Matter changes the boundary-response field and the metric; the resulting metric constrains matter trajectories and light paths. Gravity is called an anchor because the field equation is bidirectional and constraint-forming, not merely because an anchor sounds similar to gravity.

In a slowly varying region, the leading source coupling is often written G_eff≈G/λ. Where λ varies significantly, the derivative terms cannot be discarded. Those terms are precisely where a boundary gradient can generate a rotation or lensing signal. Any galactic application that keeps G/λ while dropping the radial λ derivatives at the same location is incomplete.

17. One dark mechanism in five descriptions

The historical corpus used several descriptions of the dark sector. They become coherent when placed at different levels of one causal chain rather than read as rival ontologies.

Description	Level	Meaning in the unified chain
Strategy B anchor	System organisation	A region whose prior identity has saturated into retained boundary infrastructure.
Boundary-only matter	Channel accounting	Energy that gravitates through boundary response but has no light-channel record.
Hidden curvature	Metric observation	Actual boundary/metric response not reconstructed electromagnetically as visible matter.
Adjacent recursion influence	Multiverse boundary condition	Persistent pressure and shear sourced by neighbouring loop rungs.
Second E_8 sector	Microscopic encoding proposal	A possible internal representation of electromagnetically orthogonal dark degrees of freedom.

EQUIVALENCE MAP

Adjacent recursion supplies the external boundary condition; the boundary field carries it inside our metric; Strategy B states retain it as anchor infrastructure; observers infer the metric effect as dark mass; a second E_8 sector is a proposed microscopic encoding. These are successive descriptions of one mechanism, not five independent sources of missing gravity.

The global density budget and the local radial law answer different questions. Ω_DM states how much clumping boundary content is available cosmologically. The galaxy operator states how that content and the adjacent-rung stress are distributed around a baryonic source. A correct density fraction does not by itself derive a rotation curve.

18. The gold-silver interface

Our universe is the n=1 golden attracting loop. The n=2 silver loop is its child and adjacent upper boundary. The two loops reverse orientation on each iteration but contract at different rates and spend different fractions of a complete cycle in the boundary channel.

$$\mu_1 = -1/\varphi^2 \approx -0.381966, \quad \mu_2 = -1/(1+\sqrt{2})^2 \approx -0.171573 \quad (18.1)$$

$$\Delta B_{12} = W\_B^{(1)} - W\_B^{(2)} = -2\sqrt{2} + 5/4 + 3\sqrt{5}/4 \quad (18.2)$$

$$\Delta B_{12} = 0.098623858379 \quad (18.3)$$

Plainly: the shared interface cannot remain perfectly phase-matched. During each coupled loop, a fixed fraction of golden boundary participation has no simultaneous silver boundary state. It must slip against the more light-dominant silver side. "Rubbing" is the plain-language name for this persistent unmatched boundary shear.

CONDITIONAL - BOUNDARY-SHEAR INTERPRETATION

The algebra proves the mismatch. Its interpretation as a physical inter-universe shear requires permanent adjacent-rung contact and a boundary field capable of transmitting the mismatch into our metric. Those are explicit realisation conditions, not hidden assumptions.

18.1 Why the radial shape is logarithmic

The silver eigenvalue defines multiplicative shells. Let $R_{k+1}=\varphi_2 R_k$. Equal interface work per shell means equal increments per multiplicative step. The continuous shell count is:

$$k(R) = \ln(R/R_0)/\ln\varphi_2 \quad (18.4)$$

A constant boundary increment per shell therefore accumulates as $\ln R$. Its radial derivative is proportional to $1/R$. The silver lattice fixes the outer radial shape without choosing an arbitrary halo profile.

$$\Phi_I(R) \propto \ln(R/R_0) \quad \Rightarrow \quad g_I(R) = d\Phi_I/dR \propto 1/R \quad (18.5)$$

18.2 The dimensional interface scale

ΔB_{12} is dimensionless. At the cosmological interface the available universal speed is c and the recurrence clock is H_0. Dimensional analysis leaves one minimal acceleration:

$$a_{12} = cH_0\Delta B_{12} \quad (18.6)$$

CONDITIONAL - CLOCK REALISATION

This scale follows uniquely from c, H_0, and the dimensionless boundary mismatch once the Hubble recurrence is identified as the dimensional clock. Choosing a late-time UM clock sets its numerical value; no galaxy-specific parameter is fitted.

$$H_0 = 72.1 \text{ km s}^{-1} \text{ Mpc}^{-1} \quad \Rightarrow \quad a_{12} = 6.908558055369\text{e-}11 \text{ m s}^{-2} \quad (18.7)$$

19. Galaxy rotation and apparent extra mass

Let g_b be the acceleration generated by the recorded baryonic distribution, g the total metric acceleration, and $g_I=g-g_b$ the interface contribution. The boundary is a mixed channel, so its minimal local constitutive law must be bilinear rather than an independently added mass profile. Require a homogeneous degree-two closure that vanishes when either the baryonic source or the silver mismatch vanishes, contains no new scale, recovers $g\to g_b$ at high acceleration, and allows a scale-free exterior.

$$g \cdot g\_I = a_{12}\, g\_b, \quad \text{with } g\_I = g - g\_b \quad (19.1)$$

$$g^2 - g\_bg - a_{12}g\_b = 0 \quad (19.2)$$

$$g(g\_b) = [g\_b + \sqrt{}(g\_b^2 + 4a_{12}g\_b)]/2 \quad (19.3)$$

PROPOSED - MINIMAL BILINEAR GALAXY CLOSURE

Equation (19.1) is the unique minimal bilinear closure under the stated source, scale, and limiting requirements. It is a physical constitutive proposal generated by the boundary semantics and must be judged by rotation, lensing, morphology, clusters, and the full frozen SPARC catalogue.

19.1 Limits and observable consequences

For $g\_b \gg a_{12}$, the positive root approaches the baryonic field:

$$g = g\_b + a_{12} + O(a_{12}{}^2/g\_b) \quad (19.4)$$

For $g\_b \ll a_{12}$:

$$g \approx \sqrt{}(a_{12}g\_b) \quad (19.5)$$

Outside a finite baryonic mass M_b, $g\_b = GM\_b/R^2$, so:

$$g \approx \sqrt{}(GM\_ba_{12})/R \quad (19.6)$$

$$v^4 = GM\_ba_{12} \quad (19.7)$$

The same equation gives the extra mass a Newtonian reconstruction would infer:

$$M\_app(R) = R^2[g{-}g\_b]/G \approx \sqrt{}(M\_ba_{12}/G) \cdot R \quad (19.8)$$

$$\rho\_app(R) \approx [1/(4\pi R^2)]\sqrt{}(M\_ba_{12}/G) \quad (19.9)$$

ONE RESULT, FOUR READINGS

Flat outer rotation, the baryonic fourth-power relation, linearly increasing inferred halo mass, and an effective $1/R^2$ density are not separate fitted claims. They are four observational reconstructions of the same boundary-shear closure.

19.2 Lensing

The interface contribution is not a force attached only to massive tracers. It is a metric potential. In the exterior limit its potential is:

$$\Phi\_I(R) = \sqrt{(GM\_ba_{12})} \ln(R/R_0) \quad (19.10)$$

Because the same metric potentials determine null trajectories, the logarithmic term produces an isothermal-like lensing contribution. The exact deflection and gravitational-slip relation depend on the kinetic function Z(λ), the potential U(λ), and the solution for both weak-field potentials. Rotation and lensing must therefore be fitted jointly from the same field solution; an independent lensing mass is not permitted.

20. Black holes, lensing, and recursive organisation

A galaxy halo and a black hole are the same Strategy B class at different saturation. The distributed halo is a boundary response with nonzero light reporting and an extended recovery region. A black-hole horizon is the limiting surface at which outward light reporting collapses and the boundary record saturates. The interior is not inferred by pretending the light channel still samples it normally; it is reconstructed from horizon and exterior records.

$$\lambda \to 0 \text{ at a saturated horizon, while boundary and retained content carry the record} \quad (20.1)$$

The multiverse black-hole picture is then a scale extension of the same source model. Successful attracting recursions remain organised around the primordial boundary infrastructure; failed or destabilised attempts are retained by that infrastructure. Our luminous universe is accretion-disc-like because propagation is generated and recorded at an active boundary, not because the universe is literally a thin astrophysical disk in an external Euclidean room.

SCALE-TRANSLATION RULE

Structural recurrence preserves roles, not every coordinate equation. A black hole, a galaxy, an atom, and a recursion universe can share the anchor/content/boundary architecture while retaining different local dynamics and dimensional realisations.

Part V - Cosmos, quanta, matter, and life

21. Cosmological accounting

The global density budget is a channel-accounting result. It does not replace the local field equations. The matter channel splits into a recorded baryonic projection and a binding or unrecorded projection. The boundary channel splits into clumping and smooth components through the golden survival ratio.

$$\Omega\_b = r^2/2 \quad (21.1)$$

$$\Omega\_DM = 2r(1-r)/\varphi = 4r^2(1-r) \quad (21.2)$$

$$\Omega\_DE = (1-r)^2 + 2r(1-r)(1-1/\varphi) + r^2/2 \quad (21.3)$$

$$\Omega\_b + \Omega\_DM + \Omega\_DE = 1 \quad (21.4)$$

Sector	Closed form	UM value	Grade
Baryonic matter	$r^2/2$	0.047746	CONDITIONAL accounting
Dark clumping boundary	$4r^2(1-r)$	0.263932	CONDITIONAL accounting
Smooth / dark energy remainder	$1-9r^2/2+4r^3$	0.688322	CONDITIONAL accounting

The closure is exact inside the accounting map. The assignment of each algebraic projection to a measured cosmological density remains a physical realisation and is tested against cosmological records.

21.1 The cosmological-constant exponent

The old corpus alternated between r^{240} and r^{241}. The two are reconciled by separating internal traversal from recorded closure. The E_8 root traversal contains 240 internal steps. A physical observable requires one final boundary registration. Thus:

$$\text{internal cancellation scale} \propto r^{240} \quad (21.5)$$

$$\text{recorded cosmological residual} \propto r^{241} \quad (21.6)$$

CONDITIONAL - 240 PLUS RECORD

The +1 is not another E_8 root. It is the realisation cost of making the completed internal traversal available as a physical boundary record. The distinction removes the previous notation collision while leaving both historical calculations identifiable.

22. The Hubble braiding structure and dimensional clocks

Early versions represented the late/early expansion disagreement as three traversal floors, $3r^3$. A later physical refinement included the transmission speed of the gravitational-memory channel. These are not two independent predictions. The first is the coarse traversal count; the second is the transmission-corrected realisation.

$$\text{coarse structural scale: } 3r^3 = 0.088525\ldots \quad (22.1)$$

$$\text{transmission-corrected scale: } W\_M(2/\sqrt{5}) = 0.085410\ldots \quad (22.2)$$

GOVERNING CHOICE

The final corpus uses the transmission-corrected expression for the physical Hubble split and retains $3r^3$ as the historical first-order traversal estimate.

The two observational routes are sensitive to different system histories: early-universe light propagation and late-universe matter/anchor organisation. The framework therefore treats the difference as a channel-dependent record rather than assuming that one route directly measures an identical operational object.

23. Quantum, holographic, thermodynamic, and electromagnetic sectors

The same three-role architecture appears in the quantum and statistical language. A possible state propagates, a boundary interaction selects and records a relation, and a stable material system retains the result. The boundary is the physical location of measurement, entanglement bookkeeping, and holographic encoding; it is not an observer's consciousness.

$$P\_\text{survival per golden cycle} = 1 - 1/\varphi^2 = 1/\varphi \quad (23.1)$$

This identity is proved from the two eigenmodes of the golden recurrence. Its use as a per-cycle registration coefficient is conditional on the measurement realisation. Standard squared-amplitude probabilities are not replaced merely by writing $1/\varphi$; the coefficient modifies or resolves a particular boundary-registration process that must be experimentally isolated.

Sector	Unified Mechanics reading	Status
Holography	Boundary stores the relation between retained content and outward geometric report.	Conditional realisation
Entanglement	A shared boundary record constrains separated local readouts.	Proposed mechanism
Thermodynamics	Irreversible records arise because stable forward recurrence exports unresolved content through the boundary.	Conditional
Statistical mechanics	Ensembles count admissible micro-realisations of one retained macro-description.	Conditional

Sector	Unified Mechanics reading	Status
Electromagnetism	Light-channel phase freedom and boundary registration realise gauge reporting.	Proposed/conditional
Quantisation	Discrete completed traversals carry integer multiples of the minimal action record.	Conditional

The document does not treat a dimensionless traversal floor as numerically equal to $\hbar$ in SI units. A dimensional action map must specify the clock, energy scale, and record apparatus. That map is an explicit remaining derivation, not an unexplained physical mechanism.

24. Particle and unification ledger

The particle corpus contains many r- and φ-only closed forms. In the final two-paper architecture they are gathered into a ledger rather than distributed as separate governing papers. Each entry carries three independent questions: Is the algebra exact? Is the physical identification derived? Was the comparison preregistered and executed against an independent record?

Observable	Closed form	Value	Grade / role
Fine structure inverse	$\varphi^{10}+\varphi^{5}+\varphi^{2}+\varphi^{-2}$	137.082	Conditional identification; empirical comparison
Weak mixing	$r/[2(1-r)]$	0.2250	Conditional identification
Proton/electron ratio	$\varphi^{15}(1+r)(1+r^{3})$	1838.0	Proposed mass encoding
Strong coupling at M_Z	$r^{2}(1+r-r^{2})$	0.1159	Proposed running anchor
CKM phase	$\arccos(W_B)$	1.1296 rad	Proposed boundary-phase encoding
Neutrino hierarchy	$m_e(1+r)/\varphi^{34}$ and $/\varphi^{38}$	~52.5 and 7.65 meV	Proposed mode-depth encoding
Baryon asymmetry	r^{18}	6.60×10^{-10}	Proposed three-crossing depth
Unification scale	$M_{Pl}\,\varphi^{-14}$	$\sim1.45\times10^{16}$ GeV	Conditional embedding

LEDGER RULE

A numerical match is not promoted into a mechanism. The mechanism must specify the mode count, traversal, representation, coupling, and record that select the formula. Entries that lack that chain remain proposed even when their arithmetic is precise.

24.1 Heterotic and E_8 realisation

The heterotic $E_8\times E_8$ construction is retained as a proposed microscopic realisation rather than made the logical foundation of the whole framework. In the proposed chain, the visible E_8 encodes light-coupled matter and gauge structure; the second E_8 encodes electromagnetically orthogonal boundary-sector degrees of freedom; a G_2/Fibonacci sector supplies a golden fusion eigenvalue. The PLA and system ontology do not depend on this embedding being correct.

CONJECTURED EMBEDDING

E_8 and G_2 may realise the internal channel and particle architecture. Failure of that embedding would require a different microscopic representation; it would not erase the proved projective loop algebra or the separately testable macroscopic interface law.

25. Living and semantic systems

Living systems do not require a new axiom. They are realised systems whose retained identities are actively repaired, whose local conditions are sensed, and whose records alter later admissible processes. Biological memory is therefore a material Familiarity Kernel: it retains structured records, compares them to present conditions, and changes future execution.

$$\text{life} = \text{retained identity} + \text{active boundary regulation} + \text{record-dependent reconstruction} \tag{25.1}$$

Language is a higher-order boundary system. A sentence propagates distinctions, resolves relations at grammatical and semantic boundaries, and deposits a reconstructable record in another system. The semantic word system and logic compiler model this operation. Their success is judged by reconstruction: can the receiver recover the relevant system roles and local conditions after compression?

NO ANTHROPIC EXCEPTION

Observation is downstream of system architecture. Human language is one realised recording technology, not the operation that causes the universe to have structure.

Part VI - Tests, solvers, and governance

26. What the simulations establish

The structure simulations are real computations with fixed algorithms and visible outputs. They establish implementation facts: a three-role projection can be imposed on unrelated substrates; mixed events can be tracked as emergent boundary events; and some rewriting systems approach the target channel composition more closely than others. They do not establish that the chosen substrate is the universe or that a visual resemblance is a physical proof.

Simulation family	Question answered	Question not answered
Reaction-diffusion	Can one field display propagation-, interface-, and accumulation-dominant morphologies?	Does the universe obey the selected Gray-Scott parameters?
Causal hypergraph	Do mixed light/matter rewrite events generate a distinct interface class?	Is the selected rewrite rule the fundamental physical rule?
Fractal / Eden growth	Can one channel grammar organise recursive growth across scale?	Does matching morphology determine a gravitational kernel?
CMB and cosmic web renders	Can UM-derived weights seed reproducible sky and structure visualisations?	Does a render replace a likelihood test against survey data?

SIMULATION RULE

A simulation is evidence for a mechanism only to the extent that its governing equations, initial conditions, observables, and comparison metric are fixed independently of the desired image.

27. LiMB, cosmological execution, and empirical records

LiMB is the execution layer that translates the r-derived cosmological ledger into CAMB-compatible parameters and returns CMB spectra and related observables. The current trivial-channel source limit tests whether the derived background parameters reproduce a standard Boltzmann calculation. Nonzero channel perturbations require their own source equations and cannot be claimed from background agreement alone.

Layer	Current role
Closed-form input derivations	Maps r to cosmological parameter values.
Frozen cosmology object	Prevents hidden fitting after the derivation.
CAMB backend	Executes standard perturbation evolution for those inputs.
Planck reference	Provides a direct spectrum comparison.
Channel source modules	Currently mark the path toward nontrivial L/B/M perturbations.
Test suite	Pins formulas and detects accidental drift in implementation.

The reported 101-observable suite and chain records are retained as empirical history. The final publication should freeze the exact dataset versions, likelihoods, nuisance treatment, and pass criterion used for each table. A change in the formula or data release creates a new test record rather than silently rewriting the old one.

28. SPARC: translator baseline and physical pilot

28.1 What the old test actually showed

The earlier UM-Q galaxy model used the coordinates $W\_B=2a\_L(1-a\_L)$ and $W\_M=(1-a\_L)^2$ with an intercept. Algebraically, those coordinates span the same space as $\{1,a\_L,a\_L^2\}$. The numerical equality between UM-Q and a generic quadratic therefore proved a translator-basis equivalence. It did not test a derived gravitational operator and could not nullify the physical galaxy mechanism.

$$\mathrm{span}\{1,W\_B,W\_M\} = \mathrm{span}\{1,a\_L,a\_L^2\} \quad (28.1)$$

HISTORICAL RESULT PRESERVED

The old result is now named the quadratic compiler baseline equivalence. It remains valuable because it prevents a change of basis from being advertised as new physics.

28.2 The silver-interface pilot

Model	Galaxy-balanced acceleration RMSE	Galaxy-balanced velocity RMSE
Baryons only	0.527467	43.332 km/s
Fixed RAR	0.240355	17.154 km/s
Previous UM age/light closure	0.237422	17.012 km/s
Silver-interface closure	0.223741	pilot record

The zero-fit silver closure was executed on the same reconstructed 12-galaxy, 270-point pilot. It improved the galaxy-balanced acceleration RMSE and was favoured over fixed RAR in 8 of 12 galaxies. A galaxy bootstrap remained compatible with zero difference because the pilot is small. This is evidence that the derived operator is worth a frozen full-catalogue test; it is not a completed SPARC verdict.

NEXT REQUIRED EXECUTION

Freeze stellar mass-to-light assumptions, gas sign convention, distance and inclination handling, quality cuts, covariance treatment, and every comparison law before running the full catalogue. The result must be reported whether it improves or fails.

29. Falsification surfaces

Claim	Direct falsifier or failure condition
Primitive PLA metallic spine	A counterexample to the one-inversion primitive theorem under the stated definition, or a required primitive edit not generated by A and J.
Channel partition	A demonstrated failure of the operational path mapping, or a physical process requiring a fourth primitive path under the same two-rate completion.

Claim	Direct falsifier or failure condition
Boundary gravity field	Rotation and lensing cannot be fit by one metric field without an independent dark mass component.
Silver galaxy closure	Frozen full-SPARC, lensing, cluster, or morphology tests reject the fixed law outside declared uncertainties.
Boundary-only dark record	A reproducible electromagnetic coupling intrinsic to the dark sector contradicts structural orthogonality.
Cosmological accounting	Frozen background and perturbation predictions fail jointly beyond the specified realisation uncertainties.
Born registration coefficient	A correctly isolated single-cycle boundary experiment excludes the predicted coefficient.
Heterotic embedding	The required spectrum, anomaly structure, or coupling map cannot be produced by the proposed $E_8 \times E_8/G_2$ construction.

A failure at one layer is localised. A failed E_8 embedding does not disprove the whole-part closure. A failed silver galaxy law does not disprove PLA. Conversely, a proved PLA theorem does not rescue a failed physical realisation. The dependency graph prevents both exaggerated confirmation and exaggerated nullification.

30. Final corpus theorem and closure ledger

FINAL CORPUS STATEMENT

Unified Mechanics is a physical source model, an internal mathematical reconstruction, and a family of explicit realisations. The source model begins with realised systems retaining distinctions under local conditions and leaving records. The mathematical reconstruction compresses those roles into whole, part, remainder, two complementary rates, a three-path partition, and a projective loop algebra. The realisations map propagation to Light, mixed interface to Boundary, retention to Matter, boundary response to gravity, saturated anchors to black holes, adjacent loop mismatch to dark boundary shear, and recorded cosmological and laboratory outputs to observables. No equation becomes physical without carrier, coupling, condition, and record; no observation becomes a theorem merely by matching a number.

The framework is now coherent in two senses. Historically, the physical picture is acknowledged as the origin of the mathematics. Logically, the mathematics is ordered before the physical conclusions it supports. The two orders are joined by a visible bridge rather than forced into one chronology.

30.1 Closed arguments

- **System ontology.** Realised carrier, retained identity, parts, boundary, local conditions, process, and record are separated.
- **Semantic architecture.** Normal form and physical realisation are no longer conflated.
- **Golden closure.** The positive whole-part-remainder fixed point is unique.
- **PLA.** Accumulation and inversion are complete for the admitted positive lattice edits; the full algebra is distinguished from the primitive metallic spine.
- **Genesis correction.** Substrate, repelling threshold, $n=0$ mirror, $n=1$ universe, and $n=2$ child have distinct roles.
- **Dark-sector architecture.** Anchor, boundary-only content, hidden curvature, adjacent recursion, and E_8 encoding occupy one layered chain.

- **Galaxy mechanism.** Silver mismatch, logarithmic shells, dimensional scale, bilinear closure, rotation, apparent mass, and lensing arise in one testable model.
- **Historical tests.** The quadratic translator equality is preserved and no longer misreported as a physical null.

30.2 Work that remains empirical rather than conceptually missing

- Full frozen SPARC and joint rotation-lensing execution.
- A complete disk and cluster solution of the boundary field equation.
- Nontrivial L/B/M perturbation sources in LiMB and a frozen cosmological likelihood run.
- Dimensional derivation of the action quantum from an explicit clock and energy carrier.
- Microscopic derivation or rejection of the $E_8 \times E_8 / G_2$ representation.
- Independent laboratory tests of the proposed boundary-registration and isomer couplings.

WHAT REMAINS IS TEST, NOT SILENCE

The final edition does not leave a physical claim as an unexplained number. Where a mechanism is proposed, its assumptions and record chain are stated. The remaining questions are whether nature executes those mechanisms and whether the fixed quantitative laws survive independent data.

Appendix A - Exact identities

Identity	Closed form
Golden fixed point	$\varphi^2=\varphi+1$; $1/\varphi=\varphi-1$
Contraction	$r=1/(2\varphi)=(\sqrt{5}-1)/4$
Weights	$W\_L=(1-r)^2$; $W\_B=2r(1-r)$; $W\_M=r^2$
Closure	$W\_L+W\_B+W\_M=1$
Channel ratio	$W\_L/W\_M=5$
Floor	$\varepsilon=r^3$
Mirror-to-gold crossing	$1/8-r^3=r^2$
Metallic rung	$\varphi_n=(n+\sqrt{(n^2+4)})/2$
Rung multiplier	$\mu_n=-1/\varphi_n^2$
Gold-silver mismatch	$\Delta B_{12}=-2\sqrt{2}+5/4+3\sqrt{5}/4$
Galaxy closure	$g(g-g\_b)=a_{12}g\_b$
Exterior fourth-power law	$v^4=GM\_ba_{12}$

Appendix B - Formula and status ledger

Quantity	Formula	Status
Ω_b	$r^2/2$	Conditional cosmological representation
Ω_{DM}	$4r^2(1-r)$	Conditional boundary-clumping representation
Ω_{DE}	$1-9r^2/2+4r^3$	Conditional closure
n_s	$1-r^2/\varphi^2$	Proposed primordial-spectrum mapping
Λ record	r^{241}	Conditional 240-step traversal + record
Hubble split	$W_M \cdot 2/\sqrt{5}$	Proposed transmission-corrected realisation
a_{12}	$cH_0\Delta B_{12}$	Conditional dimensional interface scale
Galaxy g	$[g_b+\sqrt{(g_b^2+4a_{12}g_b)}]/2$	Proposed fixed constitutive law
Born registration	$1/\varphi$	Conditional measurement realisation
Action floor	r^3 per complete traversal	Conditional until dimensional map is closed

Appendix C - Equivalence and provenance map

Historical phrase	Final location	Final meaning
Gravity as anchor	Boundary field equation	Bidirectional metric constraint realised by W_B.
Dark matter as Strategy B	System organisation	Distributed retained boundary infrastructure.
Dark matter as boundary-only matter	Channel accounting	Gravitating content with no direct light-channel record.
Dark matter as hidden curvature	Observation	Metric response inferred as unseen mass.
Dark matter as adjacent universes	Multiverse boundary	External condition carried by neighbouring rungs.
Dark matter as second E_8	Microscopic proposal	Possible encoding of the orthogonal dark degrees of freedom.
SPARC null	Historical translator test	Equality of two quadratic bases, not a gravity null.
$n=-1$ primordial universe	Corrected genesis	Repelling threshold separated from unclosed substrate and $n=0$ mirror.
$n=2$ parent	Corrected family relation	Silver is a child and adjacent upper neighbour, not our parent.

Appendix D - Definitions

Term	Definition
Boundary	The realised interface at which a distinction, transfer, or role reversal is resolved and may leave a record.
Carrier	The nonempty physical system in which a formal relation is realised.
Familiarity Kernel	A record-preserving classification stage for structured but not yet stably identified phenomena.
Primitive universe	A closed PLA word with no proper inversion-delimited internal loop.
Record	A persistent physical difference that can support reconstruction or inference.
Realisation map	An assignment of internal formal roles to a carrier, coupling, local condition, process, and observable.
Strategy A	Stable organisation around an anchor within a recovery basin.
Strategy B	Resolution into retained boundary infrastructure when Strategy A identity cannot be sustained.
Universopedia	The observation-first grammar that preserves carrier, condition, record, and competing inference.
Rubbing / boundary shear	Persistent unmatched boundary participation between adjacent loop rungs with different phase and contraction rates.

Appendix E - Selected references and corpus provenance

Shields, J. Unified Mechanics corpus, physical papers, simulations, mathematical monographs, and solver records (2025-2026).

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FINAL READING ROUTE

This Main Monograph is the governing account. The Compact Monograph is its strict compression. All prior papers remain part of the development archive and reproducibility record, but none overrides the definitions, status grades, or dependency order stated here.